

Partial Fractions and Improper Integrals



Partial fraction decomposition: distinct linear factors

- 1 Decompose $\frac{5x-1}{(x-1)(x+2)}$ into partial fractions.
 - 2 Evaluate $\int \frac{3x+5}{x^2+x-2} dx$ using partial fractions (note $x^2 + x - 2 = (x + 2)(x - 1)$).
 - 3 Decompose $\frac{x+7}{x^2-x-6}$ into partial fractions (note $x^2 - x - 6 = (x - 3)(x + 2)$).
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Partial fractions with a repeated linear factor

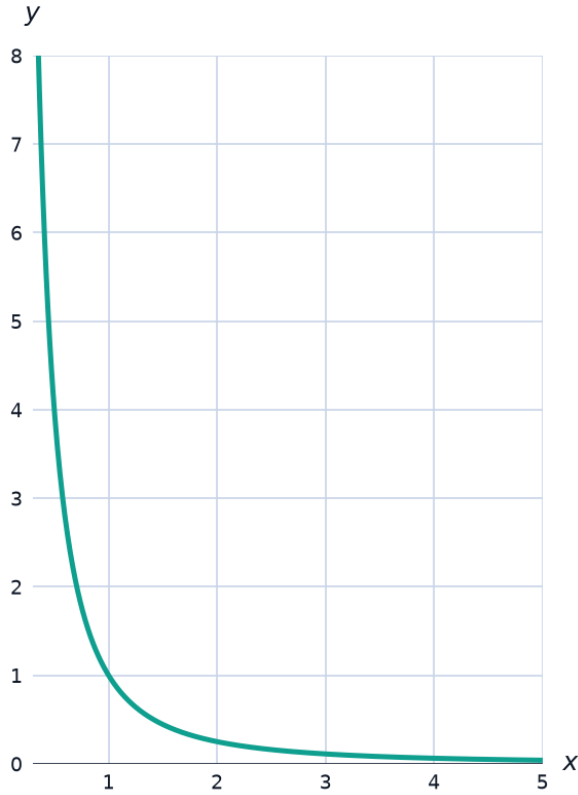
- 4 Decompose $\frac{2x+3}{(x-1)^2}$ into partial fractions.
 - 5 Evaluate $\int \frac{2x+3}{(x-1)^2} dx$ using the decomposition found above.
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Partial fractions with an irreducible quadratic factor

- 6 Set up (but do not solve for constants) the partial fraction decomposition of $\frac{3x^2+2}{(x-1)(x^2+4)}$.
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Improper integrals: infinite limits

- 7 Evaluate $\int_1^{\infty} \frac{1}{x^2} dx$, or show it diverges.



- 8 Evaluate $\int_1^{\infty} \frac{1}{x} dx$, or show it diverges.
- 9 Evaluate $\int_0^{\infty} e^{-2x} dx$, or show it diverges.
- 10 Evaluate $\int_{-\infty}^0 e^x dx$, or show it diverges.

Improper integrals: discontinuous integrands

- 11 Evaluate $\int_0^1 \frac{1}{\sqrt{x}} dx$, or show it diverges (note the integrand is undefined at $x = 0$).
- 12 Evaluate $\int_0^1 \frac{1}{x} dx$, or show it diverges.
- 13 Evaluate $\int_0^2 \frac{1}{(x-2)^2} dx$, or show it diverges (the integrand is undefined at $x = 2$).

The p-test for improper integrals

- 14 State the p-test for $\int_1^{\infty} \frac{1}{x^p} dx$, and use it to determine convergence for $p = 3$ and $p = 0.5$.

- 15 State the analogous p-test for $\int_0^1 \frac{1}{x^p} dx$ (a discontinuity at $x = 0$), and determine convergence for $p = \frac{1}{2}$ and $p = 2$.
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Combining partial fractions with improper integrals

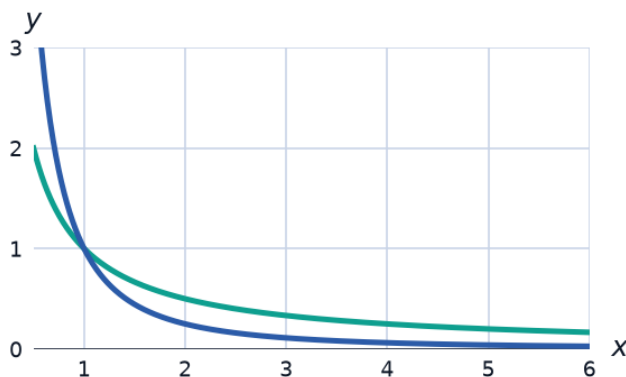
- 16 Evaluate $\int_0^\infty \frac{1}{(x+1)(x+2)} dx$ using partial fractions.
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Three distinct linear factors

- 17 Decompose $\frac{6x^2 - 3x - 1}{x(x-1)(x+1)}$ into partial fractions.
- 18 Evaluate $\int \frac{6x^2 - 3x - 1}{x(x-1)(x+1)} dx$ using the decomposition above.
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Visualizing convergence: area under the curve

- 19 The convergent improper integral $\int_1^\infty \frac{1}{x^2} dx = 1$ (shown earlier) means the infinite region under $y = \frac{1}{x^2}$ for $x \geq 1$ has FINITE area. Explain intuitively why this is possible even though the region extends forever.
- 20 By contrast, $\int_1^\infty \frac{1}{x} dx$ diverges. Compare the graphs of $y = \frac{1}{x}$ and $y = \frac{1}{x^2}$ on $[1, 10]$ and explain why $\frac{1}{x}$ decays too slowly for the area to stay finite.



Improper integrals with both an infinite bound and a discontinuity

- 21 Evaluate $\int_0^\infty \frac{1}{\sqrt{x}(1+x)} dx$ is a classic doubly-improper integral (discontinuous at $x = 0$ AND infinite upper bound). Explain how such an integral must be split to evaluate it.
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Comparison test for improper integrals

- 22 Use a direct comparison to argue that $\int_1^{\infty} \frac{1}{x^2+1} dx$ converges, without computing it exactly.
- 23 Use a direct comparison to argue that $\int_1^{\infty} \frac{2+\sin x}{x} dx$ diverges.
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Multiple choice — easy 15

- 24 For $\frac{7x+5}{(x+3)(x-4)} = \frac{A}{x+3} + \frac{B}{x-4}$, find A .

a. $-\frac{16}{7}$

b. $\frac{23}{7}$

c. $\frac{16}{7}$

d. $\frac{33}{7}$

- 25 For $\frac{2x}{(x+3)(x-3)} = \frac{A}{x+3} + \frac{B}{x-3}$, find A .

a. 1

b. -1

c. $\frac{3}{2}$

d. 2

- 26 For $\frac{7x-5}{(x)(x+5)} = \frac{A}{x} + \frac{B}{x+5}$, find A .

a. 8

b. -1

c. 1

d. 0

27 For $\frac{7x+2}{(x+6)(x+5)} = \frac{A}{x+6} + \frac{B}{x+5}$, find A .

a. 40

b. -33

c. -40

d. 41

28 For $\frac{2x-6}{(x)(x+4)} = \frac{A}{x} + \frac{B}{x+4}$, find A .

a. $\frac{7}{2}$

b. $-\frac{3}{2}$

c. $\frac{3}{2}$

d. $-\frac{1}{2}$

29 For $\frac{6x-7}{(x+6)(x-4)} = \frac{A}{x+6} + \frac{B}{x-4}$, find A .

a. $\frac{53}{10}$

b. $\frac{17}{10}$

c. $\frac{43}{10}$

d. $-\frac{43}{10}$

30 For $\frac{7x+1}{(x-3)(x-2)} = \frac{A}{x-3} + \frac{B}{x-2}$, find A .

a. 23

b. -22

c. -15

d. 22

31 For $\frac{2x-8}{(x-2)(x)} = \frac{A}{x-2} + \frac{B}{x}$, find A.

a. 2

b. -2

c. 4

d. -1

32 For $\frac{2x-7}{(x-1)(x+4)} = \frac{A}{x-1} + \frac{B}{x+4}$, find A.

a. -1

b. 1

c. 0

d. 3

33 For $\frac{6x+9}{(x+1)(x-5)} = \frac{A}{x+1} + \frac{B}{x-5}$, find A.

a. $\frac{1}{2}$

b. $\frac{13}{2}$

c. $-\frac{1}{2}$

d. -1

34 For $\frac{5x-4}{(x-3)(x+5)} = \frac{A}{x-3} + \frac{B}{x+5}$, find A.

a. $-\frac{11}{8}$

b. $\frac{29}{8}$

c. $\frac{19}{8}$

d. $\frac{11}{8}$

35 For $\frac{3x+3}{(x+3)(x-5)} = \frac{A}{x+3} + \frac{B}{x-5}$, find A.

a. $\frac{7}{4}$

b. $-\frac{3}{4}$

c. $\frac{9}{4}$

d. $\frac{3}{4}$

36 For $\frac{4x}{(x)(x-3)} = \frac{A}{x} + \frac{B}{x-3}$, find A.

a. $-\frac{1}{2}$

b. 0

c. 1

d. 4

37 For $\frac{3x-4}{(x-4)(x-5)} = \frac{A}{x-4} + \frac{B}{x-5}$, find A.

a. -8

b. 8

c. -7

d. 11

38 For $\frac{8x+9}{(x-6)(x-4)} = \frac{A}{x-6} + \frac{B}{x-4}$, find A.

a. $-\frac{57}{2}$

b. $\frac{57}{2}$

c. $-\frac{41}{2}$

d. $\frac{59}{2}$

Multiple choice — medium 15

39 Evaluate $\int_1^{\infty} \frac{2}{x^3} dx$, or show it diverges.

a. -1

b. 1

c. 2

d. $\frac{2}{3}$

40 Evaluate $\int_1^{\infty} \frac{7}{x^2} dx$, or show it diverges.

a. $\frac{7}{2}$

b. 7

c. 8

d. -7

41 Evaluate $\int_1^{\infty} \frac{9}{x^6} dx$, or show it diverges.

a. $\frac{9}{5}$

b. $\frac{3}{2}$

c. $-\frac{9}{5}$

d. $\frac{14}{5}$

42 Evaluate $\int_1^{\infty} \frac{8}{x^2} dx$, or show it diverges.

a. -8

b. 4

c. 8

d. 9

43 Evaluate $\int_1^{\infty} \frac{3}{x^3} dx$, or show it diverges.

a. $-\frac{3}{2}$

b. $\frac{3}{2}$

c. $\frac{5}{2}$

d. 1

44 Evaluate $\int_1^{\infty} \frac{3}{x^2} dx$, or show it diverges.

a. $\frac{3}{2}$

b. -3

c. 4

d. 3

45 Evaluate $\int_1^{\infty} \frac{6}{x^5} dx$, or show it diverges.

a. $-\frac{3}{2}$

b. $\frac{6}{5}$

c. $\frac{5}{2}$

d. $\frac{3}{2}$

46 Evaluate $\int_1^{\infty} \frac{4}{x^2} dx$, or show it diverges.

a. 5

b. -4

c. 2

d. 4

47 Evaluate $\int_1^{\infty} \frac{9}{x^3} dx$, or show it diverges.

a. $-\frac{9}{2}$

b. $\frac{11}{2}$

c. 3

d. $\frac{9}{2}$

48 Evaluate $\int_1^{\infty} \frac{7}{x^6} dx$, or show it diverges.

a. $\frac{7}{6}$

b. $\frac{7}{5}$

c. $\frac{12}{5}$

d. $-\frac{7}{5}$

49 Evaluate $\int_1^{\infty} \frac{2}{x^4} dx$, or show it diverges.

a. $\frac{2}{3}$

b. $\frac{5}{3}$

c. $\frac{1}{2}$

d. $-\frac{2}{3}$

50 Evaluate $\int_1^{\infty} \frac{1}{x^6} dx$, or show it diverges.

a. $-\frac{1}{5}$

b. $\frac{1}{6}$

c. $\frac{1}{5}$

d. $\frac{6}{5}$

51 Evaluate $\int_1^{\infty} \frac{8}{x^4} dx$, or show it diverges.

a. $\frac{8}{3}$

b. $-\frac{8}{3}$

c. $\frac{11}{3}$

d. 2

52 Evaluate $\int_1^{\infty} \frac{6}{x^3} dx$, or show it diverges.

a. 4

b. 3

c. 2

d. -3

53 Evaluate $\int_1^{\infty} \frac{3}{x^4} dx$, or show it diverges.

a. $\frac{3}{4}$

b. 1

c. 2

d. -1

Multiple choice — hard 15

54 Evaluate $\int_0^1 \frac{1}{x^3} dx$, or show it diverges (the integrand is undefined at $x = 0$).

a. $\frac{8}{3}$

b. $\frac{5}{2}$

c. $-\frac{5}{3}$

d. $\frac{5}{3}$

55 Evaluate $\int_0^1 \frac{2}{x^3} dx$, or show it diverges (the integrand is undefined at $x = 0$).

a. $\frac{8}{3}$

b. $\frac{11}{3}$

c. 8

d. $-\frac{8}{3}$

56 Evaluate $\int_0^1 \frac{1}{x^3} dx$, or show it diverges (the integrand is undefined at $x = 0$).

a. -4

b. 5

c. 4

d. $\frac{4}{3}$

57 Evaluate $\int_0^1 \frac{2}{x^3} dx$, or show it diverges (the integrand is undefined at $x = 0$).

a. $\frac{9}{2}$

b. -4

c. 4

d. 5

58 Evaluate $\int_0^1 \frac{6}{x^3} dx$, or show it diverges (the integrand is undefined at $x = 0$).

a. $\frac{15}{2}$

b. 30

c. $-\frac{15}{2}$

d. $\frac{17}{2}$

59 For $\frac{5x-1}{(x-3)^2} = \frac{A}{x-3} + \frac{B}{(x-3)^2}$, find B .

- a. 16
- b. 4
- c. 14
- d. -14

60 For $\frac{5x-9}{(x-4)^2} = \frac{A}{x-4} + \frac{B}{(x-4)^2}$, find B .

- a. 11
- b. -4
- c. 13
- d. -11

61 For $\frac{4x+8}{(x-5)^2} = \frac{A}{x-5} + \frac{B}{(x-5)^2}$, find B .

- a. 28
- b. 12
- c. 30
- d. -28

62 For $\frac{5x-3}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$, find B .

- a. -2
- b. 4
- c. 6
- d. 2

63 For $\frac{5x+1}{(x+3)^2} = \frac{A}{x+3} + \frac{B}{(x+3)^2}$, find B .

a. 6

b. 14

c. -14

d. -12

64 Use a direct comparison to determine whether $\int_1^{\infty} \frac{4+\sin x}{x} dx$ converges or diverges.

a. Diverges only because $a = 4$ happens to be even.

b. Diverges, since $\frac{4+\sin x}{x} \geq \frac{3}{x}$ for $x \geq 1$, and $\int_1^{\infty} \frac{3}{x} dx$ diverges since $p = 1 \leq 1$.

c. Converges, since $\frac{4+\sin x}{x} \leq \frac{5}{x}$ for $x \geq 1$, and $\int_1^{\infty} \frac{5}{x} dx$ converges since $p = 1 > 1$.

d. Cannot be determined by comparison, since $\sin x$ oscillates.

65 Use a direct comparison to determine whether $\int_1^{\infty} \frac{2+\sin x}{x^2} dx$ converges or diverges.

a. Cannot be determined by comparison, since $\sin x$ oscillates.

b. Diverges, since $\frac{2+\sin x}{x^2} \geq \frac{1}{x^2}$ for $x \geq 1$, and $\int_1^{\infty} \frac{1}{x^2} dx$ diverges since $p = 2 \leq 1$.

c. Converges, since $\frac{2+\sin x}{x^2} \leq \frac{3}{x^2}$ for $x \geq 1$, and $\int_1^{\infty} \frac{3}{x^2} dx$ converges since $p = 2 > 1$.

d. Converges only because $a = 2$ happens to be even.

66 Use a direct comparison to determine whether $\int_1^{\infty} \frac{2+\sin x}{x^3} dx$ converges or diverges.

a. Cannot be determined by comparison, since $\sin x$ oscillates.

b. Diverges, since $\frac{2+\sin x}{x^3} \geq \frac{1}{x^3}$ for $x \geq 1$, and $\int_1^{\infty} \frac{1}{x^3} dx$ diverges since $p = 3 \leq 1$.

c. Converges, since $\frac{2+\sin x}{x^3} \leq \frac{3}{x^3}$ for $x \geq 1$, and $\int_1^{\infty} \frac{3}{x^3} dx$ converges since $p = 3 > 1$.

d. Converges only because $a = 2$ happens to be even.

67 Use a direct comparison to determine whether $\int_1^{\infty} \frac{5+\sin x}{x} dx$ converges or diverges.

a. Diverges only because $a = 5$ happens to be even.

b. Diverges, since $\frac{5+\sin x}{x} \geq \frac{4}{x}$ for $x \geq 1$, and $\int_1^{\infty} \frac{4}{x} dx$ diverges since $p = 1 \leq 1$.

c. Cannot be determined by comparison, since $\sin x$ oscillates.

d. Converges, since $\frac{5+\sin x}{x} \leq \frac{6}{x}$ for $x \geq 1$, and $\int_1^{\infty} \frac{6}{x} dx$ converges since $p = 1 > 1$.

68 Use a direct comparison to determine whether $\int_1^{\infty} \frac{5+\sin x}{x^2} dx$ converges or diverges.

a. Cannot be determined by comparison, since $\sin x$ oscillates.

b. Converges only because $a = 5$ happens to be even.

c. Converges, since $\frac{5+\sin x}{x^2} \leq \frac{6}{x^2}$ for $x \geq 1$, and $\int_1^{\infty} \frac{6}{x^2} dx$ converges since $p = 2 > 1$.

d. Diverges, since $\frac{5+\sin x}{x^2} \geq \frac{4}{x^2}$ for $x \geq 1$, and $\int_1^{\infty} \frac{4}{x^2} dx$ diverges since $p = 2 \leq 1$.